

VAYUSETU: Digital Twin System Architecture

Comprehensive Technical Framework & Scientific Flow

1. Scientific Underpinnings & Core Concept

VAYUSETU is a production-grade Climate Digital Twin (CDT) platform of India. Unlike traditional weather dashboards that display static and unlinked observations, VAYUSETU establishes a continuous feedback loop between the physical Earth systems and their digital replicas. It assimilates multi-source national and regional datasets, tracks dataset lineage, propagates uncertainty indices, runs dynamic physical and policy scenario simulations, and provides explainable AI predictions coupled with multi-stakeholder decision recommendations.

2. High-Level System Architecture Flow

The platform functions as a unified pipeline structured as follows:

- **Data Ingestion Layer:** Downloads and ingests live observation feeds from ground stations, Doppler radars, and satellites including IMD, INSAT-3D/3DR, MOSDAC, Bhuvan, and NICES.
- **Data Assimilation Layer:** Implements a closed-loop 1D/2D Kalman Filter to reconcile raw observations with baseline climate model predictions, correcting state variables in real-time.
- **Digital Twin State Store:** Stores versioned, audited twin states capturing temperature, rainfall, soil moisture, humidity, LST, SST, and albedo.
- **AI Prediction Engine:** Runs multi-model spatio-temporal predictions including hybrid Convolutional LSTM (ConvLSTM), Temporal Fusion Transformers (TFT), XGBoost, and Physics-Informed Neural Networks (PINN).
- **Scenario Simulation Layer:** Calculates what-if policy impacts (forest cover, CO2 level, urbanization shift, soil moisture) and routes flood runoff using a 2D Saint-Venant hydraulic equation solver.
- **Risk Fusion & Recommendation Layer:** Fuses multiple hazards into a single Climate Resilience Index (CRI) and triggers targeted actions for farmers, administrators, and disaster management units.

3. The Closed-Loop Digital Twin Process

The platform operates on a continuous loop of **Predict** → **Observe** → **Assimilate** → **Correct** → **Predict**. When a satellite or ground telemetry pack is received (e.g. INSAT-3D Land Surface Temperature), it is compared with the AI model prediction. The Kalman Filter computes the dynamic Kalman Gain (K) to weight the prediction error, correcting the state variables before storing the final Twin State. This ensures the digital twin state matches physical reality closely, correcting for sensor noise and model drift.